

JAYESH TAK

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EDUCATION

Master of Science in Computer Software Engineering

May 2024

Northeastern University, Boston, MA

Relevant Courses: Application Design, Cloud Computing, Web Development, Data Structures & Algorithms

TECHNICAL SKILLS

Programming Languages	Java, Python, JavaScript, C/C++, C#, PHP
Web Development	HTML, CSS, Bootstrap, Angular, React.js, Next.js, Node.js, Express.js, Redux
Cloud Tools	AWS, GCP, Jenkins, GitHub Actions, Packer, Terraform, Pulumi
Tools/Framework	Spring Boot, Hibernate, ASP.NET, Laravel, Git, Jenkins, OpenCV, OpenGL
Data Management Systems	MySQL, PostgreSQL, SQLite, MongoDB, AWS AuroraDB, Elasticsearch
Other Applications/Tech	REST APIs, Linux/Unix, Gen AI, Postman, AWS Lambda, EC2, S3, Route53

WORK EXPERIENCE

BOSCH, Plymouth, MI

May 2024 – Present

Software Engineer

- Developed automation scripts with HashiCorp Packer to containerize applications in AWS AMI, achieving 99% automation for app startup during EC2 instance launches
- Optimized real-time image processing in C++ by resolving distortion with advanced masking techniques, enhancing System Visualization Software (SVS) functionality
- Designed and implemented an algorithm in Java and OpenSceneGraph for image segmentation and feature extraction, reducing processing time by 30% and improving object detection accuracy by 15%
- Built and maintained Java-based backend services for processing ECU commands, enabling seamless server integration and diagnostics for a mobile app
- Built and maintained Java-based backend services enabling real-time ECU command execution and diagnostics, integrating WebSocket protocols to improve system responsiveness and usability
- Designed and developed RESTful APIs using Spring Boot and Hibernate, enabling seamless interaction between front-end applications and databases, ensuring scalability and reliability
- Integrated front-end and back-end systems, ensuring smooth data flow and implementing secure authentication mechanisms using JWT and OAuth
- Developed a mobile application with real-time ECU command execution and diagnostics, reducing response time and enhancing system usability

Northeastern University, Boston, MA

Sep 2023 – Apr 2024

Research & Teaching Assistant

- Led hands-on sessions on AWS, utilizing EC2, S3, and Lambda to demonstrate cloud-based application hosting
- Abridged learning gaps and effectively instructed students in Java and MERN stack, along with debugging

Mantic Technologies, Bengaluru, India

Jan 2020 – Aug 2022

Software Engineer

- Led the design and development of a high-volume food delivery service application using Java, Spring Boot, and Hibernate resulting in improved scalability and boosting customer engagement by over 25%
- Teamed with DevOps to establish CI/CD pipelines using GitHub Actions, reducing deployment time by 30%
- Built RESTful APIs to handle product catalog management in Java and ORM ensuring seamless data interaction
- Developed JSON-based RESTful services in Node.js using Sequelize ORM, enhancing PostgreSQL database efficiency by 30%, validated with Swagger API and Postman
- Built an e-commerce platform including content migrations and upgrades to purchase/sell goods using ReactJS

Merizon Technologies, Hyderabad, India

Apr 2019 – Dec 2019

Jr. Software Engineer

- Developed scalable backend applications using Java and Spring Boot, reducing API response times by 30% through optimized logic, caching mechanisms, and clean coding practices
- Managed and optimized MySQL and MongoDB databases, improving query execution time by 40% using indexed tables, schema normalization, and slow query analysis
- Designed secure RESTful APIs with microservices architecture, reducing system downtime by 25% through JWT authentication, API rate-limiting, and robust testing
- Enhanced system reliability by achieving 95% test coverage with JUnit and Mockito, resolving critical bugs, and automating testing through CI/CD pipelines
- Collaborated with cross-functional teams in Agile environments, delivering projects 15% ahead of schedule with clear communication and detailed documentation

PROJECTS

Cloud Native Web Application — CI/CD, GitHub Actions, RESTAPI, AWS, PULUMI

- Developed a cloud-native backend application with NodeJS, adhering to the 12-factor app methodology

Car Rental Web Application — Spring Boot, PostgreSQL, RESTful APIs

- Developed a Java-based Full-Stack Car Rental web app using Spring Boot, integrating RESTful APIs for seamless communication; optimized indexing with RabbitMQ, Elasticsearch, Kafka, and Google OAuth 2.0; used Kibana for visualization, Docker for containerization, and achieved 95% test coverage with JUnit and API validation via Postman